

CATALYSIS

A hydrophobic FeMn@Si catalyst increases olefins from syngas by suppressing C1 by-products

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Although considerable efforts have been made in the selective conversion of syngas [carbon monoxide (CO) and hydrogen] to olefins through Fischer-Tropsch synthesis (FTS), ~50% of the converted CO is transformed into the undesired one-carbon molecule (C1) by-products [carbon dioxide (CO₂) and methane (CH₄)]. In this study, a core-shell FeMn@Si catalyst with excellent hydrophobicity was designed to hinder the formation of CO₂ and CH₄. The hydrophobic shell protected the iron carbide core from oxidation by water generated during FTS and shortened the retention of water on the catalyst surface, restraining the side reactions related to water. Furthermore, the electron transfer from manganese to iron atoms boosted olefin production and inhibited CH₄ formation. The multifunctional catalyst could suppress the total selectivity of CO₂ and CH₄ to less than 22.5% with an olefin yield of up to 36.6% at a CO conversion of 56.1%.

Olefins are feedstocks for many high-value-added chemical products and are usually produced from petroleum (1), so there is increasing interest in alternative production routes from nonpetroleum carbon resources (2), including syngas (a mixture of CO and H₂) that can be derived from biomass, coal, and natural gas (3, 4). Recently, many breakthroughs have been made in the conversion of syngas to olefins. Bao and colleagues (5, 6) reported the concept of oxide-zeolite (OX-ZEO), separating CO activation and C-C coupling on different active sites. At 400°C, ZnCrO_x/SAPO and MnO_x/SAPO exhibited a high selectivity of 80% for light olefins (C₂ to C₄) with CO conversion of ~17 and ~8%, respectively. Later, Wang and colleagues (7) prepared a ZnZrO_x/SAPO catalyst that had 70% selectivity for light olefins with ~10% CO conversion at 400°C. Although greater olefin selectivity in hydrocarbons was obtained through the OX-ZEO pathway, a lower catalytic activity and olefin yield were exhibited even at a very high reaction temperature.

Alternatively, another route, called Fischer-Tropsch to olefins (FTO), was developed for selectively converting syngas to olefins with a relatively high activity. Zhong *et al.* (8) reported that 61% selectivity for light olefins could be obtained over CoMn oxides with ~32% CO conversion. After de Jong and colleagues (9) found excellent selectivity of light olefins over iron-based catalysts for Fischer-Tropsch synthesis (FTS), research on Fe active metal ramped up (10–12). Liu *et al.* (13) prepared

Mn-modified Fe₃O₄ nanoparticles, which exhibited 60% selectivity for light olefins with ~42% CO conversion. Our previous work (14) indicated that a core-shell Fe₃O₄@MnO₂ catalyst presented a greater olefin selectivity of 79% with ~68% CO conversion.

The olefin selectivity reported above refers to the hydrocarbon products and does not include the CO₂ fraction. However, during the conversion process of syngas, the water-gas shift (WGS) reaction is inevitable given the coexistence of CO and H₂O under high reaction temperatures, which results in a high CO₂ selectivity (between 40 and 50%) when converting CO (5–9, 14, 15). Moreover, ~10 to 20% selectivity of by-product CH₄ is generally presented in the FTS reaction, especially for the iron-based catalysts (9, 10, 13, 16). Thus, during the syngas-to-olefin reaction, >45% of the converted CO is transformed into the undesired one-carbon molecule (C1) by-products (CO₂ and CH₄) either through the OX-ZEO process (5) or the FTO path (8, 9), as depicted in Fig. 1A. Recently, some work has been done to try to address this issue (17, 18), but the total selectivity of C1 products was still very high and the olefin yield was very low. Notably, Wang *et al.* (19) and Xie *et al.* (20) reported the suppression of C1 products by a specific iron carbide phase and by cobalt promoted with Mn/Na/S, respectively. However, these catalysts could only run at relatively low CO conversion levels (<20%), implying poor olefin yield.

We present a core-shell FeMn@Si catalyst with excellent hydrophobicity that hindered the WGS reaction, which enhanced the C-C coupling reaction and inhibited the formation of CO₂ and CH₄. The chain growth probability was increased to 0.73 (fig. S1), and the selectivity for undesired C1 by-products was suppressed to 22.1% (Fig. 1A). Moreover, the multifunctional catalyst exhibited an olefin selectivity of 83.3% (including 68.2%

of α -olefins) and an olefins/paraffins molar ratio of 5.0 in the C₂₊ hydrocarbons (Fig. 1B). Reaction conditions, such as reaction temperature, space velocity, and pressure, were tuned to modify the catalytic performance (figs. S2 to S4). Under the optimum reaction conditions of 320°C, 3.0 MPa, 4000 ml·h⁻¹·g⁻¹, H₂/CO = 2, the yield of olefins reached as high as 36.6% at a CO conversion of 56.1% in a single run (table S1).

The stability of catalysts is another crucial index to advance industrial applications. In this study, the CO conversion and product distribution remained stable with time on stream (Fig. 1C). The selectivity of undesired C1 by-products was suppressed to ~23% and the olefin selectivity in total products remained at ~65% during the 100-hour stability test. All of the above catalytic tests suggested that our catalyst worked well for efficiently converting syngas to olefins while successfully suppressing the formation of undesired CO₂ and CH₄.

We used several characterization methods to study the origin of the low C1 selectivity and high olefin yield. An unpromoted Fe₂O₃@SiO₂ catalyst was used to investigate the formation path and inhibition mechanism of CO₂ during iron-based FTS reaction. An appropriate amount of Mn promoter was then added to enhance the C-C coupling reaction, which suppressed the CH₄ selectivity and boosted olefin production while maintaining low CO₂ selectivity. The core-shell Fe₂O₃@SiO₂ (Fe@Si) was prepared using the modified Stöber method with nanorod-shaped Fe₂O₃ (figs. S5 and S6). Transmission electron microscopy (TEM) images directly confirmed that the nanorod-shaped Fe₂O₃ was encapsulated by a homogeneous SiO₂ shell (Fig. 2A and fig. S7). As shown in the Fourier transform infrared (FTIR) spectra (Fig. 2C), the absorption bands at 465 and 542 cm⁻¹ were assigned to the vibration of Fe–O bonds in the Fe₂O₃ core (21), and the bands at 471, 800, and 1095 cm⁻¹ to the vibration of Si–O–Si bonds in the SiO₂ shell. Additionally, an absorption band appeared at 950 cm⁻¹, corresponding to the stretching vibration of Si–OH (22). The OH stretch of isolated silanol groups at 3740 cm⁻¹ on Fe@Si was detected in the diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) (fig. S8), providing the possibility to introduce hydrophobic –CH₃ groups through the silylation reaction (fig. S9).

The obtained Fe@Si was rendered hydrophobic (Fe@Si-c) by further organic modification with chlorotrimethylsilane (TMCS). This modification had no obvious influence on the morphology, phase composition, or pore structure of Fe@Si (figs. S10 to S12). TEM images of Fe@Si-c showed that the nanorod-shaped Fe₂O₃ was also well encapsulated by the amorphous SiO₂ shell (Fig. 2B). FTIR and DRIFTS revealed new absorption bands at ~2925 cm⁻¹ for Fe@Si-c compared with Fe@Si (Fig. 2C and fig. S8), which were attributed

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to the vibration of introduced $-\text{CH}_3$ groups (22). The water-droplet contact angle tests were also conducted to characterize the hydrophobicity of Fe@Si and Fe@Si-c. Fe@Si exhibited a contact angle of 40.2° , implying a hydrophilic surface (Fig. 2E). After the hydrophobic modification, the contact angle increased to 137.6° over the Fe@Si-c surface, indicating that hydrophilic Fe@Si had been completely converted to the hydrophobic surface of Fe@Si-c through TMCS modification. Although Fe@Si was well dispersed in water, it was difficult to infiltrate the hydrophobic surface of Fe@Si-c with water, and it formed a long-lasting separate phase (Fig. 2F). The water vapor adsorption and desorption tests also confirmed that the hydrophobic modification hindered the adsorption of water molecules (fig. S13).

The hydrophobic shell coated on iron species played a critical role in tuning the active iron phases during FTS. Compared with fresh Fe@Si, new absorption bands at 578 and 700 cm^{-1} were detected in the FTIR spectra of spent Fe@Si (Fig. 2D) and were related to the vibration of the Fe-O bond in the Fe_3O_4 phase (23); bands attributed to the vibration of the Fe-O bond in the Fe_2O_3 phase disappeared. Furthermore, both Fe_3O_4 and iron carbides were detected in the x-ray diffraction (XRD) pattern of spent Fe@Si (Fig. 3A), which was consistent with the Fe_2O_3 phase in traditional iron-based catalysts being converted into a mixture of iron oxides and carbides with reaction time on stream (4, 24, 25).

For Fe@Si-c, no absorption bands related to the vibration of Fe-O bonds in Fe_2O_3 and Fe_3O_4 were seen in the FTIR spectra after reaction (Fig. 2D), implying that iron oxides disappeared during FTS. Only iron carbides were detected in the XRD pattern of spent Fe@Si-c (Fig. 3A), demonstrating that iron oxides were completely converted to carbides during FTS. We also used Mössbauer spectra to determine the content of the iron phases. The spent Fe@Si consisted of 60.7% Fe_3O_4 and 39.3% iron carbides ($\chi\text{-Fe}_5\text{C}_2$ and $\theta\text{-Fe}_3\text{C}$), whereas the spent Fe@Si-c consisted only of iron carbides (Fig. 3, B and C, and table S2), directly confirming that the hydrophobic shell could effectively tune iron phases during reaction.

To study the phase transformation in more detail, we synthesized iron carbides by introducing a carburization procedure (320°C , 0.1 MPa, $4000\text{ ml}\cdot\text{h}^{-1}\cdot\text{g}^{-1}$, $\text{H}_2/\text{CO} = 1$, 5 hours) and compared phase composition of catalysts during FTS. After carburization, the Fe_2O_3 phase in both Fe@Si and Fe@Si-c was completely transformed into iron carbides (Fig. 3D, fig. S14, and table S3). During the subsequent FTS reaction, iron carbides inside Fe@Si were converted into a mixture of iron carbides and Fe_3O_4 phases, whereas Fe@Si-c was only com-

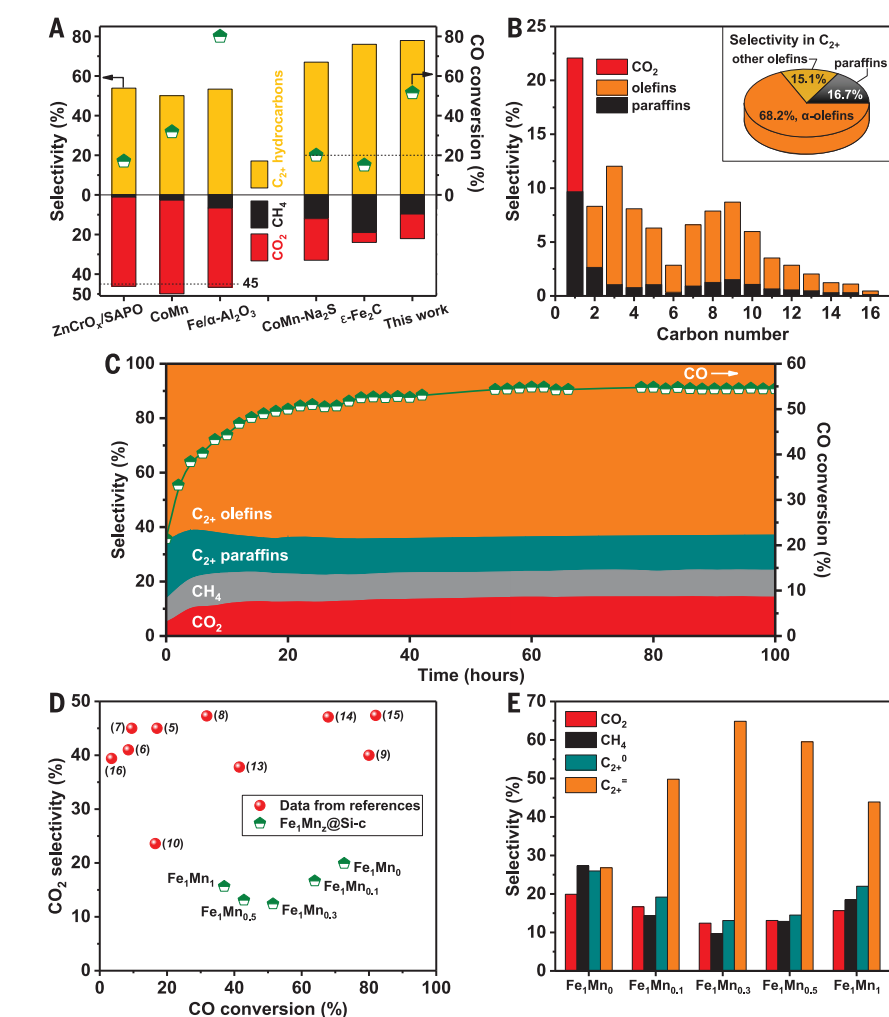


Fig. 1. Catalytic performance for syngas to olefins. (A) Product distribution (including CO_2) of FeMn@Si-c in comparison with previous work (5, 8, 9, 19, 20). **(B)** Detailed product distribution (including CO_2) and the selectivity of olefins in C_{2+} hydrocarbons (inset) over FeMn@Si-c. **(C)** A stability test of FeMn@Si-c. **(D)** CO conversion and CO_2 selectivity and **(E)** product distribution (including CO_2) over Fe₁Mn₂@Si-c with different Mn/Fe ratios. Reaction conditions: 320°C , 2.0 MPa, $4000\text{ ml}\cdot\text{h}^{-1}\cdot\text{g}^{-1}$, $\text{H}_2/\text{CO} = 2$.

posed of iron carbides (Fig. 3D, fig. S15, and table S4). Given that the only difference between Fe@Si and Fe@Si-c was their water-droplet contact angle, the different phase evolution processes on the two catalysts could be attributed to the hydrophobicity of the catalyst surface. As the FTS reaction proceeded, water was produced and the surrounding ambient became more oxidizing. As a result, iron carbides were partially converted into Fe_3O_4 by water oxidation (26–28), leading to the co-existence of iron oxides and carbides in Fe@Si. The hydrophobic surface on Fe@Si-c would facilitate rapid water desorption and inhibit its entry into the catalyst. Thus, although the FTS reaction produced water, its restricted entry inhibited the conversion of iron carbides back to oxides.

It is generally accepted that CO_2 formation on iron-based FTS catalysts consists of pri-

mary CO_2 and secondary CO_2 (19, 29–32). In the CO_2 selectivity–CO conversion curve, the extrapolated CO_2 selectivity toward the axis intercept was ascribed to the primary CO_2 ($\text{CO}^* + \text{O}^* \rightarrow 1\text{ CO}_2$, where the asterisk means chemisorbed). The increasing CO_2 selectivity above the axis intercept was assigned to the secondary CO_2 , which was related to the re-adsorption of water and its subsequent WGS reaction with CO. The slope of the curve reflected the rate of secondary CO_2 formation stemming from the WGS reaction. The CO_2 selectivity–CO conversion curve (Fig. 4A) was obtained by adjusting the flow rate at a constant H_2/CO ratio. Fe@Si exhibited a primary CO_2 selectivity of $\sim 10\%$, whereas Fe@Si-c was free of primary CO_2 . Because Fe@Si was composed of a mixture of Fe_3O_4 and iron carbides, whereas Fe@Si-c only consisted of iron carbides during reaction (Fig. 3 and fig. S16), we

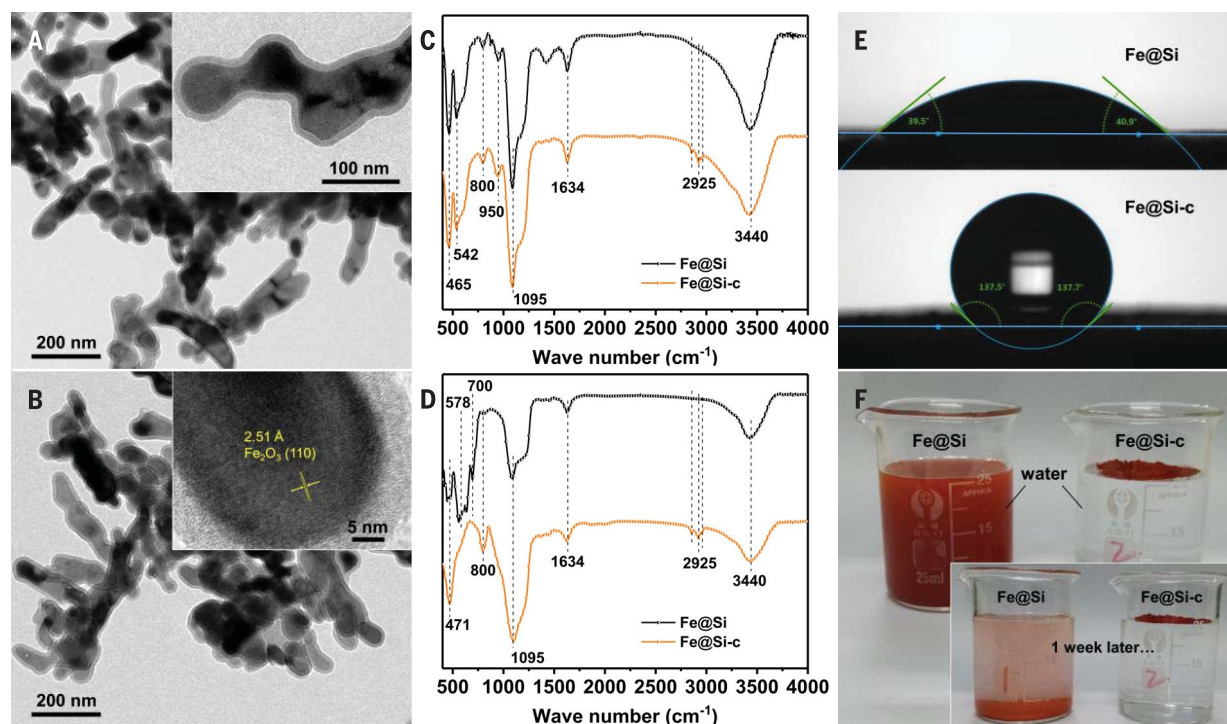


Fig. 2. Structural characterization of the Fe@Si and Fe@Si-c catalysts. (A and B) TEM images of Fe@Si (A) and Fe@Si-c (B). (C and D) FTIR spectra of the fresh (C) and spent (D) catalysts. (E) Water-droplet contact angle tests. (F) Photographs of catalysts dispersed in water, initially and 1 week later.

attributed primary CO₂ formation to the Fe₃O₄ presented in the Fe@Si catalyst, consistent with the previous report (19).

The slope of the curve obtained over Fe@Si was 1.81 times that obtained over Fe@Si-c (Fig. 4A), suggesting that Fe@Si favored the formation of secondary CO₂. However, the hydrophobic shell coated on Fe@Si-c facilitated the timely desorption of water and prevented the readsorption of water, reducing the local water partial pressure in the vicinity of iron species and inhibiting the WGS reaction. Also, the Fe₃O₄ phase is the most active phase for the WGS reaction in the iron-based FTS catalysts (19, 29, 33, 34) and would form more by-product CO₂ from CO (figs. S17 and S18 and table S5). The hydrophobic shell coated on Fe@Si-c inhibited the reoxygenation of iron carbides toward Fe₃O₄, further inhibiting the WGS reaction. A >40% reduction in CO₂ emission was achieved at a medium CO conversion level (~41%) (Fig. 4B).

To certify the above conclusions, the Fe@Si-c catalysts with different hydrophobic abilities were prepared by changing the TMCS coverage. As shown in fig. S19, the increase in the TMCS exposure increased the water contact angle from 40.2° for Fe@Si to 125.9° for Fe@Si-c_{0.5} (0.5 ml per gram of catalyst) and to 137.6° for Fe@Si-c_{2.0}, implying gradually enhanced hydrophobicity. Moreover, the slope of curve obtained over Fe@Si-c_{0.5} was 1.26 times that obtained over Fe@Si-c_{2.0} (fig. S20), further confirming that the enhancement of

catalyst hydrophobicity could suppress the WGS reaction by inhibiting the readsorption of water. For TMCS exposures >2.0 ml/g, the water contact angle and the slope of the curve tended to remain stable (figs. S19 and S20), suggesting that excessive addition of TMCS had no obvious effect on the catalyst hydrophobicity and the suppression of the WGS reaction. Fe@Si-c_{2.0} was chosen as the appropriate catalyst for the work that follows unless otherwise noted.

We tuned the H₂/CO ratio in syngas to modify the CO₂ selectivity of Fe@Si-c. As shown in fig. S21A, upon increasing the H₂/CO ratio in syngas from 1 to 4, CO conversion gradually increased from 55.5 to 80.2%, accompanied by a decrease in CO₂ selectivity from 35.0 to 15.0%, demonstrating that a higher H₂/CO ratio increased the amount of H₂ in the system and suppressed the WGS reaction. However, the selectivity of CH₄ (the other undesired C1 by-product) in hydrocarbons increased from 14.1 to 47.6% because of enhanced hydrogenation (fig. S21B).

The formation path and inhibition mechanism of CO₂, on the basis of the above results, were illustrated in fig. S22. During the FTS reaction, the adsorbed H₂* and CO* on the surface of iron species reacted to form surface CH_x species, and further C-C coupling, dehydrogenation, and hydrogenation reactions would produce various olefins and paraffins. The chemisorbed O* formed by CO dissociation could be removed by reacting with co-

adsorbed H* and CO* to form H₂O and CO₂, respectively, as oxygen carriers through primary FTS steps. Readsorption of H₂O on the surface of iron species and reaction with CO* would form secondary CO₂ through the WGS reaction. Phase transformation of iron oxides to iron carbides would remove O* from the iron oxides and form primary CO₂. During reaction, Fe@Si-c consisted only of iron carbides, which were free of primary CO₂. The inhibited adsorption of H₂O and the absence of Fe₃O₄ further suppressed the WGS reaction, so the CO₂ selectivity over the Fe@Si-c catalyst during FTS was obviously reduced. Inhibition of the WGS reaction on Fe@Si-c would decrease the amount of H₂ produced and suppress over-hydrogenation reaction, promoting olefins over CH₄ (fig. S23). The olefin selectivity in hydrocarbons increased from 11.1% of Fe@Si to 65.3% of Fe@Si-c.

Mn as a promoter is widely used to boost olefin formation in FTS. We prepared and characterized the Mn-modified catalysts (figs. S24 to S26). Fe@Si-c with moderate addition of Mn (Mn/Fe ratio ≤ 0.3) still suppressed water adsorption and inhibited the oxidation of iron carbides during reaction. However, as the Mn/Fe ratio exceeded 0.3, both Fe₃O₄ and iron carbide phases were detected in the spent Fe₁Mn_x@Si-c catalyst (figs. S27 and S28 and table S6), because excessive Mn hindered the effective contact between the hydrophobic shell and iron carbide core (fig. S29) and led to partial oxidation.

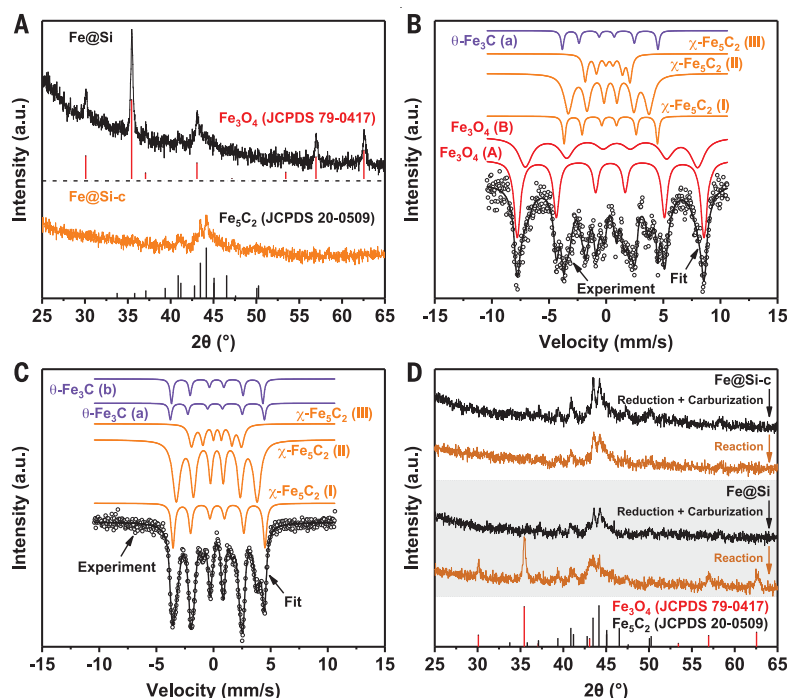


Fig. 3. Comparison of phase composition. (A) XRD patterns of the spent catalysts. (B and C) Mössbauer spectra of the spent Fe@Si (B) and Fe@Si-c (C). (D) XRD patterns of the catalysts after reduction and carburization and subsequent reaction. Reaction conditions: 320°C, 2.0 MPa, H₂/CO = 1. a.u., arbitrary units.

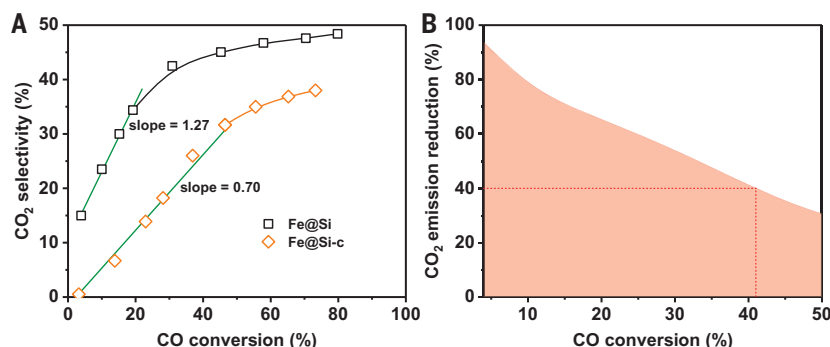


Fig. 4. Effect of hydrophobic modification on the performance of catalysts. (A) CO₂ selectivity–CO conversion curve. Reaction conditions: 320°C, 2.0 MPa, H₂/CO = 1. (B) CO₂ emission reduction of Fe@Si-c in comparison with Fe@Si. A >40% reduction in CO₂ emission was achieved at a medium CO conversion level (~41%).

The catalytic performance of Fe₁Mn_{0.3}@Si-c catalysts was shown in Fig. 1, D and E. The CO₂ selectivity obtained over all the Fe₁Mn_{0.3}@Si-c catalysts was suppressed to a low level (<20%) (Fig. 1D and fig. S30), much lower than the reported data under all the CO conversion levels in the previous references (5–10, 13–16), suggesting that the WGS reaction was still inhibited after the addition of Mn. Notably, an inverse volcano curve between CO₂ selectivity and CO conversion was observed over the Fe₁Mn_{0.3}@Si-c catalysts. CO₂ selectivity gradually decreased with the increase of Mn content and reached the minimum value of 12.4% with 51.5% CO conversion over Fe₁Mn_{0.3}@Si-c.

More importantly, moderate addition of Mn promoted the formation of olefins and suppressed that of CH₄. The olefin selectivity in total products increased with the increasing Mn content and reached the maximum value of 64.9% over Fe₁Mn_{0.3}@Si-c (Fig. 1E), which we attributed to the enhanced electron transfer from Mn to Fe atoms (fig. S31). This process facilitated the dissociated adsorption of CO and weakened hydrogenation in favor of the C–C coupling reaction and olefin production (14). Further increase of Mn content and formation of Fe₃O₄ during FTS hindered the effective contact between the hydrophobic shell and the iron species core (fig. S29), which

unproductively enhanced the WGS reaction and promoted the formation of C1 by-products (Fig. 1E and fig. S32).

REFERENCES AND NOTES

- H. M. Torres Galvis, K. P. de Jong, *ACS Catal.* **3**, 2130–2149 (2013).
- W. Zhou et al., *Chem. Soc. Rev.* **48**, 3193–3228 (2019).
- Q. H. Zhang, J. C. Kang, Y. Wang, *ChemCatChem* **2**, 1030–1058 (2010).
- J. Bao, G. H. Yang, Y. Yoneyama, N. Tsubaki, *ACS Catal.* **9**, 3026–3053 (2019).
- F. Jiao et al., *Science* **351**, 1065–1068 (2016).
- Y. F. Zhu et al., *ACS Catal.* **7**, 2800–2804 (2017).
- K. Cheng et al., *Angew. Chem. Int. Ed.* **55**, 4725–4728 (2016).
- L. Zhong et al., *Nature* **538**, 84–87 (2016).
- H. M. Torres Galvis et al., *Science* **335**, 835–838 (2012).
- J. Z. Lu et al., *ACS Catal.* **4**, 613–621 (2014).
- P. Zhai et al., *Angew. Chem. Int. Ed.* **55**, 9902–9907 (2016).
- X. Chen, D. Deng, X. Pan, Y. Hu, X. Bao, *Chem. Commun.* **51**, 217–220 (2015).
- Y. Liu, J. F. Chen, J. Bao, Y. Zhang, *ACS Catal.* **5**, 3905–3909 (2015).
- J. Wang et al., *ACS Appl. Mater. Interfaces* **10**, 43578–43587 (2018).
- V. P. Santos et al., *Nat. Commun.* **6**, 6451 (2015).
- O. Zhuo et al., *Chem. Sci.* **10**, 6083–6090 (2019).
- E. Rytter et al., *Catal. Today* **299**, 20–27 (2018).
- X. F. Yu et al., *Appl. Catal. B* **232**, 420–428 (2018).
- P. Wang et al., *Sci. Adv.* **4**, eaau2947 (2018).
- J. Xie et al., *Nat. Commun.* **10**, 167 (2019).
- S. M. Rodolfo-Baechler et al., *Mater. Lett.* **58**, 2447–2450 (2004).
- L. H. Shi, D. B. Li, B. Hou, Y. H. Sun, *Chin. J. Catal.* **28**, 999–1002 (2007).
- Y.-S. Li, J. S. Church, A. L. Woodhead, *J. Magn. Magn. Mater.* **324**, 1543–1550 (2012).
- E. de Smit, B. M. Weckhuysen, *Chem. Soc. Rev.* **37**, 2758–2781 (2008).
- M. Y. Ding et al., *Appl. Energy* **160**, 982–989 (2015).
- C. N. Satterfield, R. T. Hanlon, S. E. Tung, Z. M. Zou, G. C. Papaefthymiou, *Ind. Eng. Chem. Prod. Res. Dev.* **25**, 407–414 (1986).
- D. B. Bukur, B. Todir, N. Elbasher, *Catal. Today* **275**, 66–75 (2016).
- J. H. Liu et al., *Ind. Eng. Chem. Res.* **57**, 9120–9126 (2018).
- Y. Cheng et al., *Appl. Catal. B* **204**, 475–485 (2017).
- S. Krishnamoorthy, A. Li, E. Iglesia, *Catal. Lett.* **80**, 77–86 (2002).
- S. Li, S. Krishnamoorthy, A. Li, G. D. Meitzner, E. Iglesia, *J. Catal.* **206**, 202–217 (2002).
- M. Ojeda et al., *J. Catal.* **272**, 287–297 (2010).
- C. Pirola, C. L. Bianchi, A. Di Michele, S. Vitali, V. Ragaini, *Catal. Commun.* **10**, 823–827 (2009).
- G. P. van der Laan, A. A. C. M. Beenackers, *Appl. Catal. A Gen.* **193**, 39–53 (2000).

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SUPPLEMENTARY MATERIALS

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Materials and Methods
Figs. S1 to S32
Tables S1 to S6
References (35–39)

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